

Optimization of DNA extraction in the spiny spider crab *Maja brachydactyla*: determining the best extraction method, sample conservation and starting tissue

Elba Rodríguez-Pena¹ · Patricia Verísimo² · Luis Fernández² · Ana González-Tizón¹ · Andrés Martínez-Lage¹

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Abstract Methodological studies are an important previous step in molecular research. We optimize a method for DNA extraction from eggs and a noninvasive extraction method for adult tissues of *Maja brachydactyla*. Our results show that lyophilized eggs in B developmental stage are the most suitable for DNA extraction. Regarding adult tissues, we propose setae as an alternative tissue for DNA extraction, since prevents mutilation of specimens, which avoids infections and fitness reduction. That is especially relevant in studies with endangered species of crustaceans.

Keyword Crab · *Maja* · DNA extraction method · Egg developmental stage · Sample conservation · Sampling without death

Methodological studies focused on improving DNA extraction and sample storage (Bitencourt et al. 2007; Palero et al. 2010) are an important previous step in multiple molecular studies. In the present study we optimized the DNA extraction process from embryonic and adult tissues of *Maja brachydactyla* (Balss 1922), a decapod crustacean present in the northeastern Atlantic (Neumann 1998; Sotelo et al. 2009). We analyzed different extraction methods, sample conservation, type of adult tissue, and the developmental

stage of the eggs. We consider the three macroscopically distinguishable stages described by González-Gurriarán et al. (1998): A (orange eggs, much yolk), B (brown eggs, visible eyes) and C (dark eggs, many chromatophores, hardly yolk).

We tested several DNA extraction methodologies in fresh and stored in ethanol eggs of ten females from Golfo Ártabro and Ría de Arousa (Galicia, NW Spain). These methods included incubation in 10% bleach, heat shock, mechanical methods (cutting, puncture or homogenization), extraction with Chelex 100 (Bio-Rad), hypotonic shock and extraction with the NucleoSpin Tissue kit (Macherey–Nagel). DNA was quantified using the NanoDrop 1000 Spectrophotometer (Thermo Fisher Scientific) and we amplified the 16S gene to check its quality. We used the primers 16L29-F and 16HLeu-R (Schubart 2009), and the PCR was performed with the NZYTaq 2× Green Master Mix kit, separate MgCl₂ (Real Laboratory) in a volume of 25 µL. Each reaction contained 25 ng of DNA, 12.5 µL of Green Master Mix, 4 mM of MgCl₂ and a concentration of 0.5 µM for every primer. The PCR consisted in a denaturation of 5 min at 95 °C, 35 cycles of 30 s at 95 °C, 30 s at 53.1 °C and 45 s at 72 °C, and a final extension of 5 min at 72 °C. The amplification success was tested by electrophoresis and a gel of genomic DNA was performed to corroborate the data obtained by spectrophotometry. We only achieve to amplify the target gene with the DNA obtained by mechanical homogenization and using the commercial kit, as well as with some combinations of these methods with the others. However, the kit had the highest success rate.

We also tested the best developmental stage of eggs and sampled conservation method for DNA extraction in six *M. brachydactyla* females from the Ría de Arousa. We took two ovigerous females for each egg stage and, for each

✉ Andrés Martínez-Lage
andres@udc.es

¹ Departamento de Biología Celular y Molecular, Facultad de Ciencias de la Universidade da Coruña, Rúa da Fraga (Campus Zapateira) 10, 15008 A Coruña, Spain

² Departamento de Biología Animal, Biología Vegetal y Ecología, Facultad de Ciencias de la Universidade da Coruña, Rúa da Fraga (Campus Zapateira) 10, 15008 A Coruña, Spain

spawn, we analyzed two replicas of frozen eggs, two of lyophilized eggs and two of eggs conserved in absolute ethanol. The DNA of these eggs was extracted using the NucleoSpin Tissue kit. Concentration and quality were tested by spectrophotometry and electrophoresis. The statistical analysis showed that B stage eggs provide the higher DNA concentration. However, the ANOVA only detected statistically significant differences between the amount of DNA obtained from lyophilized and frozen eggs, being lyophilized eggs which provided more DNA.

We also took samples of pleopod setae, pereopod setae and an endite of nine females from Golfo Ártabro. The DNA of these samples was extracted using the NucleoSpin Tissue kit, and the DNA concentration and quality was checked as in the test of the conservation method. The ANOVA showed that endites and pleopod setae provide a higher DNA concentration than pereopod setae. However, we achieved to amplify the target gene from the three tissues.

Although we expected C stage to be the best developmental stage for DNA extraction, we found B stage to provide more DNA. A possible reason is the presence of substances that interfere in the extraction or hard structures that obstruct the access to the DNA in C stage eggs.

We expected freezing to be the most suitable conservation method for the posterior DNA extraction. Nevertheless, we got a higher quantity of DNA from lyophilized eggs and eggs in ethanol. One possible reason is that part of the DNA was lost during egg handling. Lyophilization and conservation in ethanol induce tissue hardening which facilitate handling. Another advantage is that storage in ethanol does not require special equipment, and DNA quality is similar that in the case of lyophilized eggs.

The most used tissue for DNA extraction in crustaceans is the muscle (McKeown and Shaw 2008; Baggio et al. 2011). However, this study aims to find alternative tissues which avoid the specimen death. This enables to continue researching with the same specimens or even the specimens' liberation after the sampling. Moreover, DNA extraction from structures that prevent the specimens' mutilation, reduces the probability of infections caused by the sampling and avoid fitness decrease. Therefore, we propose setae as an alternative tissue for DNA extraction. In females, the best tissue is indisputably the pleopod setae. Nevertheless, males pleopods are transformed into copulatory appendages, so DNA cannot be extracted from pleopod setae. Furthermore, the extraction of endites from a live specimen is complicated. So, pereopod setae seem to be a good alternative for DNA extraction in males. From these structures we obtained enough DNA to amplify and sequence the 16S gene of *M. brachydactyla*, which indicates a good quality

of the DNA. These setae have a sensorial function (Laverack 1988) and play a role in the camouflage (Fürböck and Patzner 2005). Nevertheless, sampling would involve such a small area that hardly affects crabs life. Thereby, the utilization of setae for DNA extraction enables to return the specimens to the marine environment. This is especially relevant for sampling in endangered species of crustaceans, since it allows obtaining genetic information without affecting natural populations.

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